

Panel MS-AR(1): common trend, random intercepts, catch-up

Seasonally adjusted real GDP per worker (period-average USD)

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes. The pooled and random-effects specifications are

$$y_{it} = a_i + g t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}). \quad (3)$$

In the pooled specification $a_i \equiv a$. In the random-effects specification $a_i \sim \mathcal{N}(\alpha, \omega^2)$; α and ω are estimated by maximum likelihood, integrating each country's Hamilton-filter likelihood by 11-point Gauss–Hermite quadrature. Cycles then use the posterior mean of a_i . Specifications 3 and 4 keep a permanent random intercept and add a decaying entry gap

$$y_{it} = a_i + g t + b_i \lambda^{t-T_{i0}} + z_{it}, \quad (4)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2), \quad b_i \sim \mathcal{N}(0, \omega_b^2) \text{ independent.} \quad (5)$$

T_{i0} is country i 's first observation. Spec 3 holds $\lambda = 0.98$ (Barro 2% per year). Spec 4 estimates λ in $(0, 0.985)$. The two random effects are integrated by 7×7 Gauss–Hermite quadrature. The outcome is log seasonally adjusted real GDP per worker (period-average USD) from the 2026-09-17 quarterly panel. Calendar time is taken from the period stamp t (year-fraction). g is per year and common. Latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ and ρ switch with the regime. The median μ is pinned at 0. $|\rho| < 0.99$. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters. π is the ergodic distribution of Π . $E[z]$ is the long-run mean of the cycle implied by μ, ρ , and Π .

2 Common intercept and linear trend

2026-09-17 SA panel, full sample: log real GDP per worker. 74 countries, 8670 observations, 1950-Q1–2026-Q2. Fitted countries: 74. Observations: 8670. Log-likelihood: 22607.90. Ergodic mean of the cycle $E[z] = 0.7539$ (0.0689). Common intercept $a = -6.0145$ (0.0592). Common slope $g = 0.0079$ (0.0005).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.2212 (0.0943)	0.0000 —	1.7476 (0.0464)
σ	0.0695 (0.0019)	0.0161 (0.0004)	0.0090 (0.0002)
ρ	0.9900 (0.0000)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.7511 (0.0229)	0.1700 (0.0206)	0.0789 (0.0115)
Π_1	0.0610 (0.0063)	0.9337 (0.0064)	0.0053 (0.0022)
Π_2	0.0279 (0.0035)	0.0029 (0.0016)	0.9692 (0.0034)
π	0.1487 (0.0120)	0.4010 (0.0301)	0.4502 (0.0318)

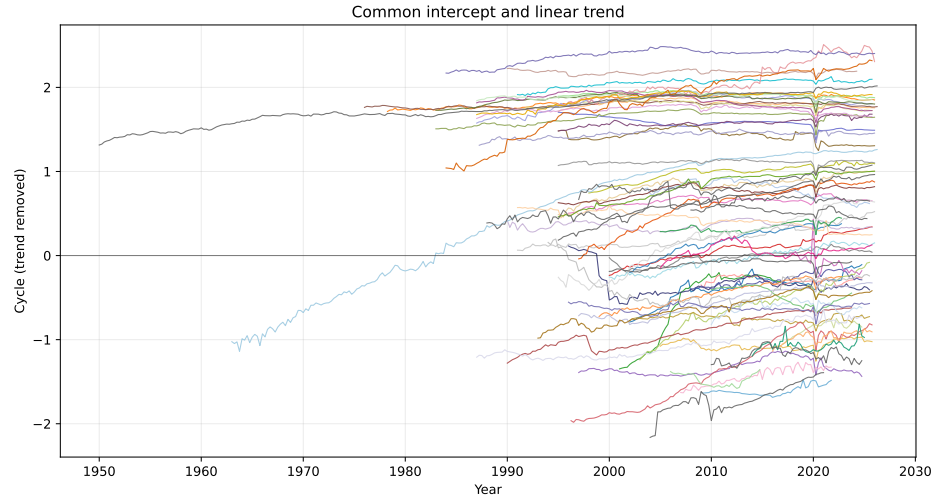


Figure 1: Country cycles after removing the linear trend.

3 Random-effects intercepts, common linear trend

2026-09-17 SA panel, full sample: log real GDP per worker. 74 countries, 8670 observations, 1950-Q1–2026-Q2. Fitted countries: 74. Observations: 8670. Log-likelihood: 23465.89. Ergodic mean of the cycle $E[z] = 0.0559$ (0.0268). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -5.9377$ (0.0424), $\omega = 0.5121$ (0.0061). Posterior-mean a_i : mean -5.536, min -7.953, max -3.922. Common slope $g = 0.0086$ (0.0005).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.8646 (0.1094)	0.0000 —	0.2813 (0.0354)
σ	0.0750 (0.0026)	0.0188 (0.0005)	0.0070 (0.0001)
ρ	0.9899 (0.0004)	0.9899 (0.0001)	0.9898 (0.0002)
Π_0	0.7023 (0.0319)	0.2780 (0.0326)	0.0197 (0.0091)
Π_1	0.0580 (0.0072)	0.8821 (0.0103)	0.0599 (0.0071)
Π_2	0.0081 (0.0029)	0.0474 (0.0062)	0.9445 (0.0055)
π	0.0949 (0.0098)	0.4190 (0.0211)	0.4861 (0.0247)

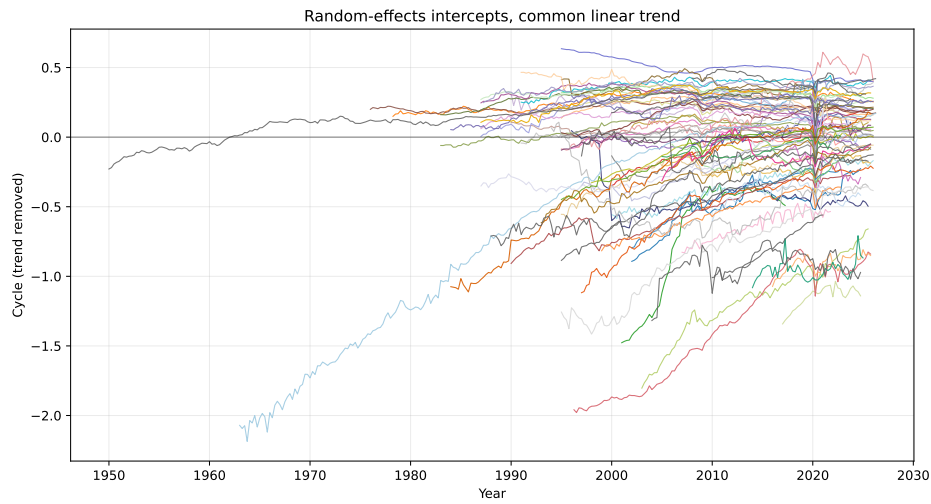


Figure 2: Country cycles after removing the linear trend.

4 RE intercepts plus catch-up, lambda fixed at Barro 0.98

2026-09-17 SA panel, full sample: log real GDP per worker. 74 countries, 8670 observations, 1950-Q1–2026-Q2. Fitted countries: 74. Observations: 8670. Log-likelihood: 23563.79. Ergodic mean of the cycle $E[z] = 0.0601$ (0.0275). Permanent RE intercepts and catch-up $y_{it} = a_i + gt + b_i\lambda^{t-T_{i0}} + z_{it}$ with $\alpha = -5.7753$ (0.0388), $\omega = 0.6254$ (0.0064), $\lambda = 0.9800$ (fixed), $\omega_b = 0.4180$ (0.0097). Posterior-mean a_i : mean -5.787, min -8.120, max -4.292. Posterior-mean b_i : mean -0.014, min -0.977, max 0.989. Common slope $g = 0.0117$ (0.0005).

Table 3: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.9844 (0.1493)	0.0000 —	0.3062 (0.0292)
σ	0.0760 (0.0028)	0.0187 (0.0005)	0.0069 (0.0001)
ρ	0.9899 (0.0007)	0.9900 (0.0001)	0.9898 (0.0008)
Π_0	0.6929 (0.0328)	0.2871 (0.0332)	0.0199 (0.0091)
Π_1	0.0543 (0.0067)	0.8882 (0.0095)	0.0575 (0.0066)
Π_2	0.0091 (0.0026)	0.0455 (0.0057)	0.9454 (0.0052)
π	0.0899 (0.0095)	0.4275 (0.0214)	0.4826 (0.0248)

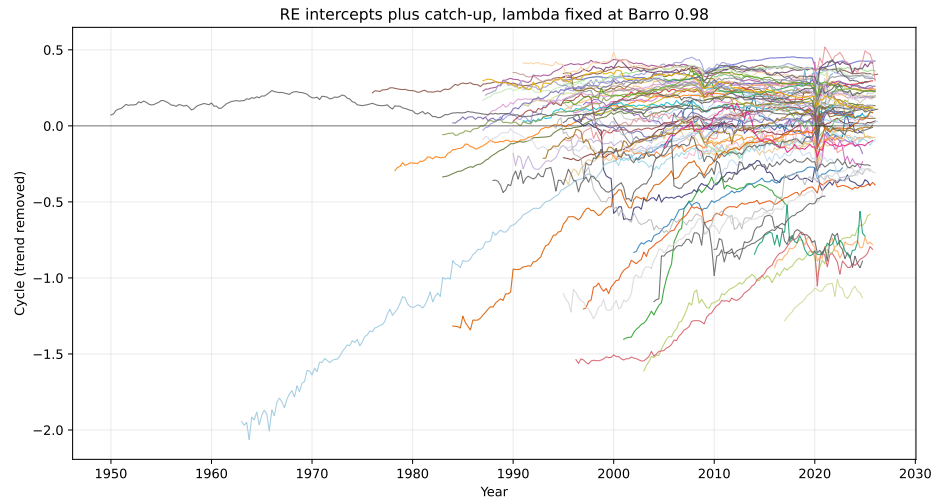


Figure 3: Country cycles after removing $a_i + gt + b_i\lambda^{t-T_{i0}}$ (posterior-mean a_i, b_i).

5 RE intercepts plus catch-up, free lambda in (0, 0.985)

2026-09-17 SA panel, full sample: log real GDP per worker. 74 countries, 8670 observations, 1950-Q1–2026-Q2. Fitted countries: 74. Observations: 8670. Log-likelihood: 23584.47. Ergodic mean of the cycle $E[z] = 0.0236$ (0.0296). Permanent RE intercepts and catch-up $y_{it} = a_i + gt + b_i\lambda^{t-T_{i0}} + z_{it}$ with $\alpha = -5.3357$ (0.0470), $\omega = 0.7218$ (0.0084), $\lambda = 0.9836$ (0.0014), $\omega_b = 0.4311$ (0.0166). Posterior-mean a_i : mean -5.464, min -8.013, max -3.627. Posterior-mean b_i : mean -0.128, min -1.016, max 1.020. Common slope $g = 0.0089$ (0.0005).

Table 4: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.9498 (0.1280)	0.0000 —	0.2469 (0.0357)
σ	0.0740 (0.0027)	0.0182 (0.0005)	0.0068 (0.0002)
ρ	0.9896 (0.0014)	0.9899 (0.0002)	0.9900 (0.0000)
Π_0	0.6927 (0.0342)	0.2953 (0.0353)	0.0120 (0.0076)
Π_1	0.0632 (0.0091)	0.8803 (0.0122)	0.0565 (0.0067)
Π_2	0.0033 (0.0047)	0.0508 (0.0082)	0.9459 (0.0056)
π	0.0940 (0.0101)	0.4327 (0.0216)	0.4732 (0.0257)

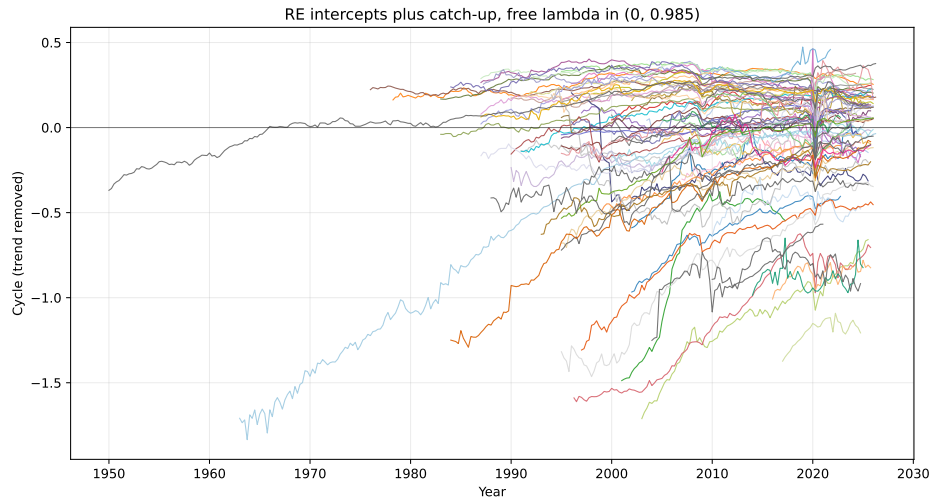


Figure 4: Country cycles after removing the deterministic path.